

```
ax.margins(0.1)
pl.scatter(xs, ys)
pl.scatter(center[0], center[1])

circle = pl.Circle(center, radius, color = 'b', fill = False)

ax.add_artist(circle)
pl.savefig(str(thename) + '.png')
pl.close()
```

---

*[Listing 7-2](#): Our improved `plot_triangle()` function, which plots the circumcenter and circumcircle*

We start by adding two new arguments: a `centers` variable that's a list of the respective circumcenters of all triangles, and a `radii` variable that's a list of the radius of every triangle's circumcircle. Note that we take arguments that consist of lists, since this function is meant to draw multiple triangles instead of just one triangle. We'll use `pylab`'s circle-drawing capabilities to draw the circles. Later, we'll be working with multiple triangles at the same time. It will be useful to have a plotting function that can plot multiple triangles instead of just one. We'll put a loop in our plotting function that will loop through every triangle and center and plot each of them successively.

We can call this function with a list of triangles that we define:

---

```
triangle1 = points_to_triangle((0.1,0.1),(0.3,0.6),(0.5,0.2))
center1,radius1 = triangle_to_circumcenter(triangle1)
triangle2 = points_to_triangle((0.8,0.1),(0.7,0.5),(0.8,0.9))
center2,radius2 = triangle_to_circumcenter(triangle2)
plot_triangle([triangle1,triangle2],[center1,center2],[radius1,radius2],'two')
```

---

Our output is shown in [Figure 7-7](#).